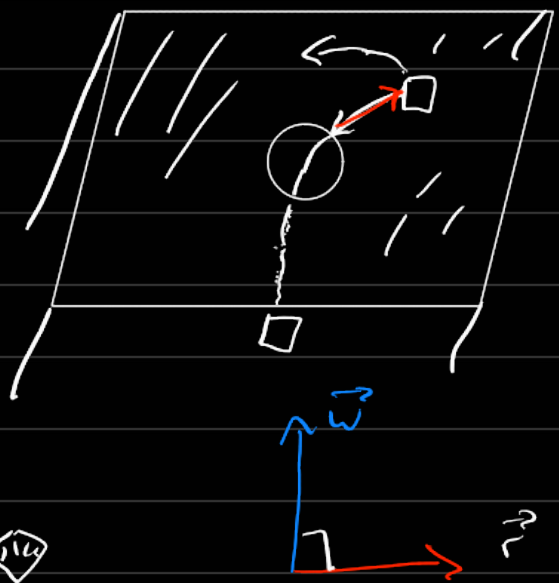
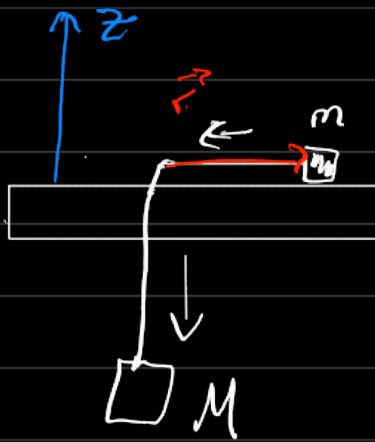
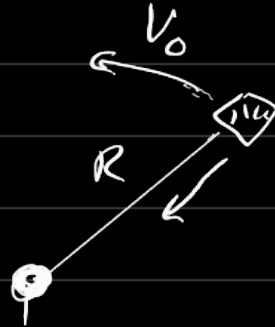


PS6 | ①

Side:



$$M = V_0$$



(angular momentum)

$$(1) \quad \vec{L} = \vec{r} \times \vec{p} = m \vec{r} \times \vec{v}$$

$$\vec{\omega} = \omega \hat{e}_z$$

$$\vec{p} = m \vec{v} = m \vec{r} \times \vec{\omega}$$

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{r} \cdot \vec{\omega} = 0$$

$$\vec{L} = m \vec{\omega} (\vec{r}^2) - m \vec{r} (\vec{r} \cdot \vec{\omega})$$

$$\begin{cases} |\vec{v}| = |\vec{\omega}| |\vec{r}| \\ v(t) = \omega r(t) \\ \omega(t) = \frac{v(t)}{r(t)} \end{cases}$$

$$\vec{L} = m |\vec{r}|^2 \vec{\omega}$$

$$\vec{L} = m |\vec{r}|^2 \omega \hat{e}_z$$

$$|\vec{L}| = m |\vec{r}|^2 \omega$$

$$|\vec{L}| = m (r(t))^2 \omega(t)$$

$$= m (r(t))^2 \left(\frac{v(t)}{r(t)} \right)$$

$$|\vec{L}| = m r(t) v(t) \quad *$$

$$\vec{r} = x(t) \hat{e}_x + y(t) \hat{e}_y$$

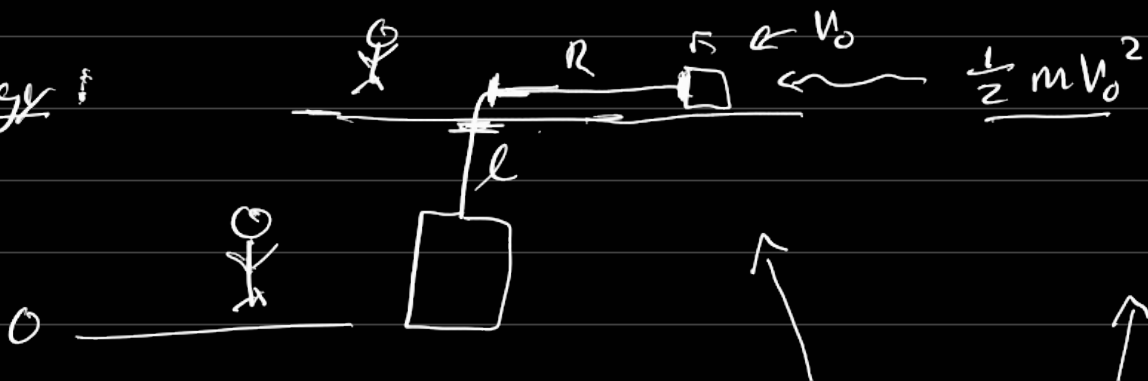
$$|\vec{r}|^2 = x^2 + y^2$$

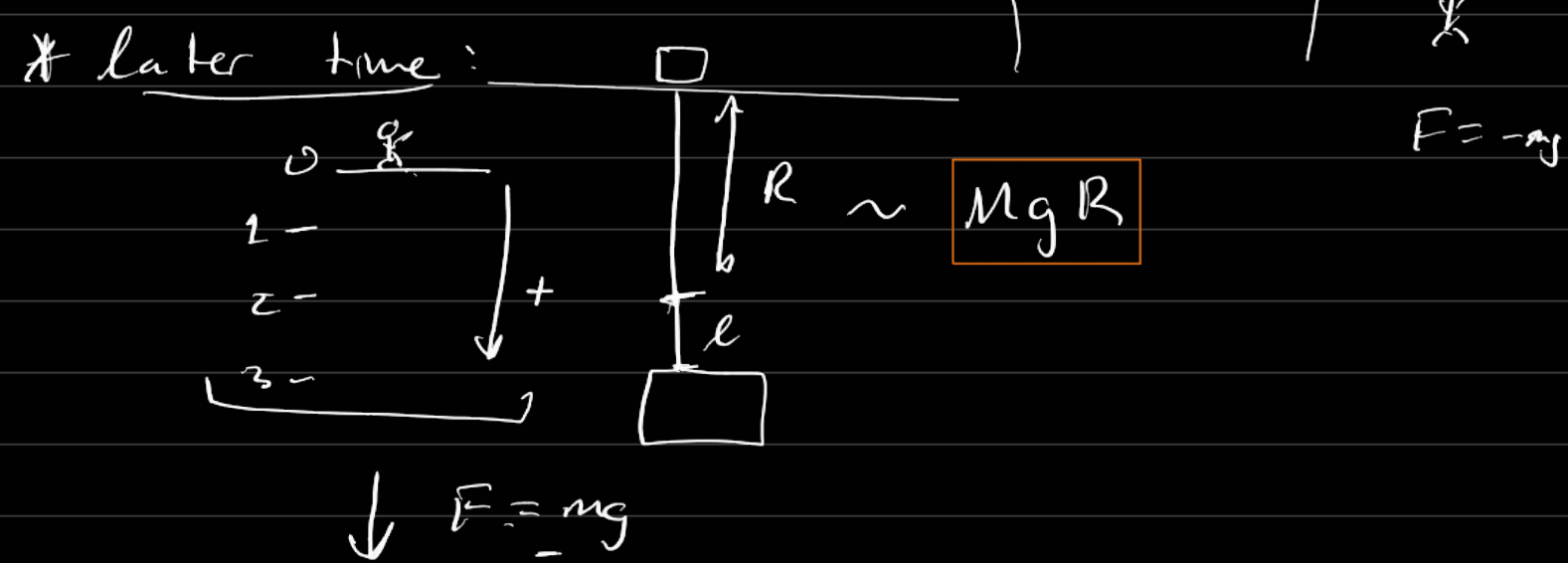
$$\begin{cases} x = r(t) \cos(\varphi) \\ y = r(t) \sin(\varphi) \end{cases}$$

$$|\vec{r}|^2 = r^2 \cos^2(\varphi) + r^2 \sin^2(\varphi)$$

$$= r(t)^2$$

* Energy:





* But, by conservation of energy

$$E = \frac{1}{2} m v_0^2 + MgR$$

* $\omega(t) = v(t) / r(t)$ $\omega(t) = \frac{d\phi}{dt} = \dot{\phi}$

$\leadsto$ K.E. $= \frac{1}{2} m v^2 = \frac{1}{2} m \dot{r}^2$

$\leadsto$ Motion from angular part

(K.E.) angular $= \frac{1}{2} m \tilde{v}^2 = \frac{1}{2} m (\dot{\phi} r)^2$

$\tilde{v} = \dot{\phi} r$

$(K.E.)_{\text{total}} = \frac{m}{2} (\dot{r}^2 + \dot{\phi}^2 r^2)$ $\swarrow$ also get

$\vec{r}(t) = x(t) \hat{e}_x + y(t) \hat{e}_y$

$\vec{r}(t) = () \hat{e}_r + () \hat{e}_\phi$

lets find $\vec{r}(t)$ in (x, y)

$\vec{r}(t) = r(t) \cos(\phi(t)) \hat{e}_x + r(t) \sin(\phi(t)) \hat{e}_y$

$\frac{d}{dt} (r(t) \cos(\phi(t))) = \dot{r}(t) \cos(\phi(t)) - r(t) \sin(\phi(t)) \dot{\phi}$

$\frac{d}{dt} (r(t) \sin(\phi(t))) = \dot{r}(t) \sin(\phi(t)) + r(t) \cos(\phi(t)) \dot{\phi}$

$\dot{\vec{r}}(t) = \dot{r}(t) (\cos(\phi) \hat{e}_x + \sin(\phi) \hat{e}_y)$

$$+ r(t) \varphi (-\sin(\varphi) \hat{e}_x + \cos(\varphi) \hat{e}_y)$$

$$\begin{pmatrix} \cos(\varphi) \\ \sin(\varphi) \end{pmatrix} = \hat{e}_r$$

$$\begin{pmatrix} -\sin(\varphi) \\ \cos(\varphi) \end{pmatrix} = \hat{e}_\varphi$$

WRONG

$$|\dot{\vec{r}}|^2 = \dot{r}^2 + r^2 \dot{\varphi}^2$$

$$\dot{r}^2 = \dot{\vec{r}}^2 + r^2 \dot{\varphi}^2 \quad | \quad r^2 \dot{\varphi}^2 = 0$$

$$\dot{\vec{r}}(t) = \dot{r}(t) \hat{e}_r + r(t) \dot{\varphi}(t) \hat{e}_\varphi$$

$$|\dot{\vec{r}}(t)|^2 = |\dot{\vec{r}}|^2 = \dot{r}^2 + r^2 \dot{\varphi}^2$$

$$K.E. = \frac{1}{2} m |\dot{\vec{r}}|^2 = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\varphi}^2)$$

Can get rid of $\dot{\varphi}$:

• Conservation of angular momentum: $|\vec{L}|$: constant

$$|\vec{L}| = m r^2 \dot{\varphi} \quad \Rightarrow \quad \dot{\varphi} = \frac{|\vec{L}|}{m r^2}$$

$$(K.E.) = \frac{1}{2} m \left(\dot{r}^2 + r^2 \left(\frac{|\vec{L}|^2}{m^2 r^4} \right) \right)$$

$$(K.E.) = \frac{1}{2} m \left(\dot{r}^2 + \frac{|\vec{L}|^2}{m^2 r^2} \right)$$

$$|\vec{L}|_{t_i} = m(r(t_i) v(t_i)) = |\vec{L}|_{t_f} = m(r(t_f) v(t_f))$$

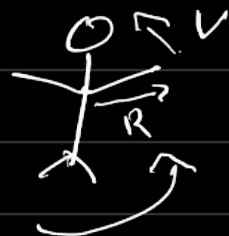
$$|\vec{L}|_{t=0} = m r(0) v(0) \leftarrow V_0$$

$$|\vec{L}| = m R V_0 = m r v$$

$$R V_0 = r \hat{v}$$

$$\hat{v} = \frac{R V_0}{r}$$

$$|\vec{L}| = m R V_0$$



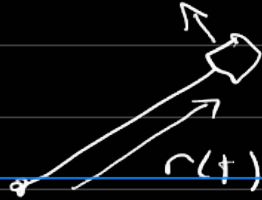
$$T(t) = \frac{1}{2} m \left(\dot{r}(t)^2 + \frac{R^2 v_0^2}{r(t)^2} \right)$$

$$r(0) = R$$

$$T(0) = \frac{1}{2} m v_0^2$$

$$v = R \dot{\varphi}$$

$\dot{r}$ that we've used $t_0 = \dot{\varphi}$



$$\dot{r}(0) = 0$$

$$T(t) = \frac{1}{2} m \left(\underbrace{\dot{r}^2}_{\text{Radial}} + \underbrace{\frac{R^2 v_0^2}{r(t)^2}}_{\text{angular}} \right)$$

